

VERKHOFYANSK, Russian Federation

WMO: 242660

Lat: 67.565N

Lon: 133.412E

Elev: 136

StdP: 99.70

Time Zone: 10.00 (E10)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-55.6	-54.1	-57.1	0.0	-54.6	-55.1	0.0	-53.4	3.6	-39.8	3.2	-41.5	0.6	240	0.796

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	28.9	18.0	26.5	16.8	24.1	15.7	18.8	27.4	17.6	25.0	16.2	22.9	2.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.5	11.1	22.4	14.1	10.2	21.8	12.7	9.3	20.2	54.3	27.4	50.1	25.1	46.0	23.0	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.4	4.5	-56.3	32.1	2.2	2.4	-57.8	33.8	-59.1	35.3	-60.3	36.6	-61.9	38.4		
(5)			-56.4	20.4	2.1	2.1	-57.9	21.9	-59.2	23.1	-60.4	24.3	-62.0	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-14.2	-45.3	-41.8	-28.3	-11.7	3.4	13.1	15.6	12.0	2.8	-14.4	-34.3	-43.2
	DBStd	23.49	6.31	6.42	8.62	7.64	5.60	4.84	4.87	4.19	4.36	9.04	8.22	6.70
	HDD10.0	9231	1716	1450	1187	651	213	25	9	26	218	757	1329	1651
	HDD18.3	11914	1974	1683	1446	901	463	167	112	199	466	1015	1579	1909
	CDD10.0	399	0	0	0	0	8	117	183	89	3	0	0	0
	CDD18.3	42	0	0	0	0	0	9	28	4	0	0	0	0
	CDH23.3	652	0	0	0	0	2	173	393	83	0	0	0	0
	CDH26.7	187	0	0	0	0	0	42	128	17	0	0	0	0
Wind	WSAvg	1.4	0.7	0.6	0.8	1.6	2.4	2.5	2.2	1.7	1.7	1.3	0.8	0.8
Precipitation	PrecAvg	180	6	6	5	6	13	29	34	29	19	13	11	8
	PrecMax	287	17	22	14	21	46	91	115	69	54	38	40	31
	PrecMin	107	0	0	0	0	0	3	4	5	2	2	3	0
	PrecStd	45	3	4	3	4	10	18	22	14	13	8	8	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-21.7	-15.2	0.6	9.0	22.2	30.5	32.9	29.1	19.4	5.3	-9.0	-19.1
		MCWB	-21.9	-16.1	-2.3	3.8	12.7	18.8	20.3	18.0	11.7	1.5	-10.2	-19.5
	2%	DB	-28.5	-23.9	-4.1	5.6	18.5	27.6	30.0	25.7	15.3	1.8	-15.5	-24.6
		MCWB	-28.6	-24.1	-6.3	1.5	10.4	16.9	19.0	16.4	9.4	-0.9	-16.0	-24.8
	5%	DB	-32.8	-28.9	-8.4	3.2	15.7	25.2	27.7	22.9	12.7	-0.1	-18.7	-29.5
		MCWB	-32.9	-29.0	-10.0	-0.2	8.8	15.2	17.8	15.2	7.9	-1.9	-19.0	-29.7
	10%	DB	-36.4	-32.7	-13.4	0.8	13.1	22.5	25.4	20.4	10.1	-2.0	-21.9	-33.2
		MCWB	-36.5	-32.8	-14.3	-2.0	7.3	14.1	16.7	13.8	6.3	-3.6	-22.1	-33.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-21.9	-16.1	-2.3	3.9	12.9	19.1	21.1	19.1	12.2	1.8	-10.2	-19.3
		MCDB	-21.7	-15.2	0.4	8.8	21.8	29.0	30.7	27.5	18.2	4.6	-9.0	-19.0
	2%	WB	-28.6	-24.1	-6.2	1.8	10.8	17.5	19.6	17.2	10.0	-0.4	-15.9	-24.8
		MCDB	-28.5	-23.9	-4.1	5.3	17.6	26.7	28.6	24.0	14.5	1.5	-15.5	-24.6
	5%	WB	-32.9	-29.0	-9.9	-0.1	9.1	15.9	18.4	15.8	8.2	-1.9	-19.0	-29.6
		MCDB	-32.8	-28.9	-8.4	3.1	15.2	23.8	26.2	21.7	12.1	-0.2	-18.6	-29.5
	10%	WB	-36.5	-32.8	-14.3	-1.9	7.5	14.5	17.2	14.4	6.6	-3.6	-22.1	-33.3
		MCDB	-36.4	-32.7	-13.4	0.9	12.9	21.7	24.2	19.6	9.7	-2.0	-21.9	-33.2
Mean Daily Temperature Range	5% DB	MDBR	5.3	8.6	17.0	16.8	10.7	11.3	11.4	11.5	9.5	9.0	6.6	5.5
		MCDBR	10.5	12.4	18.2	14.0	15.2	16.2	16.5	16.5	14.2	8.6	9.7	12.0
		MCWBR	10.4	12.2	16.2	10.6	8.8	7.7	7.3	8.2	8.6	6.6	9.3	11.8
	5% WB	MCDBR	10.5	12.4	18.0	13.3	14.0	15.0	14.6	14.5	13.0	8.0	9.8	12.0
		MCWBR	10.4	12.2	16.1	10.2	8.3	7.5	7.0	7.3	8.1	6.3	9.4	11.8
Clear-Sky Solar Irradiance	taub	0.173	0.226	0.287	0.337	0.379	0.377	0.469	0.420	0.305	0.243	0.178	N/A	
	taud	1.887	2.060	2.057	2.015	2.182	2.369	2.065	2.180	2.479	2.213	1.909	N/A	
	Ebn at Noon	152	598	737	795	806	825	711	708	753	575	126	0	
	Edh at Noon	11	49	90	126	120	103	136	109	62	41	9	0	
All-Sky Solar Radiation	RadAvg	0.05	0.70	2.41	4.68	5.46	5.67	5.14	3.84	2.10	0.84	0.12	0.00	
	RadStd	0.01	0.04	0.13	0.20	0.37	0.51	0.31	0.32	0.15	0.10	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.45	+0.58	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page